Reproducibility Report: Blair, Cooper, Coppock and Humphreys (2019)

Declaring and Diagnosing Research Designs

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This repository holds the actively maintained replication code for Blair, Cooper, Coppock and Humphreys (2019), together with the reproducibility report that documents what the original

archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1017/S0003055419000194
Replication archive	10.7910/DVN/XYT1VB

The deposit is not redistributed here. It lives at Harvard Dataverse and that is the only copy this repository points at. `download_original.R` fetches it, unpacks it, and verifies it; `verify_original.R` holds the checks and runs both before and after a full reproduction.

The deposit is a single zip file, so a check that `original/` contains nothing the manifest does not list is a check on one file and says nothing about what is inside it. The repository therefore carries two manifests. `original_manifest.csv` records the archive file itself: its Dataverse file identifier, its byte size, and two checksums, the MD5 of the bytes Dataverse serves for `?format=original` and the MD5 Dataverse publishes. `original_members_manifest.csv` records the size and MD5 of every one of the 620 members inside it. The zip is unpacked into `original_extracted/`, which `download_original.R` rebuilds from scratch on every run and nothing else writes to. Both directories are absent from version control and both are verified by size and checksum, so the exact bytes are pinned even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it, and also runs the deposited code and records what happens. This README is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains, so nothing in the chain is more restrictive than the archive itself. See `LICENSE`.

To reproduce. Clone or download the repository, open `blair_cooper_coppock_humphreys_2019.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposited archive from Dataverse, verifies it, and produces every table and figure into `maintained/output/`, then rebuilds the ground truth and re-verifies the deposit. Every diagnosis uses the published 5000 simulations and 5000 bootstrap replicates. Individual scripts can be run on their own in a fresh session, in any order, with two exceptions: `figure_2_bs_sampling_distributions.R` reads the simulations written by `table_a19_bs_diagnosis.R`, and `figure_1_experimental_designs.R` must not run in a session where the QCA appendix table has already run. That second constraint is asserted in the script rather than left to the order of the driver, and the reason is under What Moved Underneath the Code below.

Required packages: DeclareDesign, DesignLibrary, estimatr, QCA, Matching, tidyverse, future, readxl, broom, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check**, and the CSV output should come back byte-identical. The figure PDFs always show as changed, because a PDF records the time it was written; compare their PNG twins instead.

1 Paper Overview

Citation: Blair, G., Cooper, J., Coppock, A. and Humphreys, M. (2019). Declaring and diagnosing research designs. *American Political Science Review*, 113(3), 838-859.

Research question: Can a research design, quantitative or qualitative, be written down formally enough that its properties can be computed before the study is run, and what does that formalization reveal that informal design description hides?

Design: The paper is methodological rather than empirical. It proposes that a design is fully described by four elements, a model M, an inquiry I, a data strategy D and an answer strategy A, and that once all four are declared in code, any diagnosand (bias, RMSE, power, coverage, an exaggeration ratio, a posterior shift) is a simulation away. The argument is carried by two kinds of evidence. First, a census of 30 existing design tools against a single blocked working example, showing which elements of that design each tool can express. Second, a set of worked declarations spanning survey sampling, Bayesian descriptive inference, process tracing, qualitative comparative analysis, nested mixed methods, observational regression, matching, regression discontinuity, factorial and three-arm experiments, and discovery research, each diagnosed by Monte Carlo simulation. The paper closes with a design replication of Björkman and Svensson (2009), diagnosing the original study's properties without using its data.

Main finding: Formal declaration exposes design properties that are invisible otherwise. No surveyed tool can declare a correlation between block sizes and effects, an estimand, a sampling strategy, an assignment procedure or probability weighting, and none can report bias, coverage or the standard deviation of estimates. In the design replication, an estimator that appears unbiased for child mortality is shown to exaggerate the effect it detects by roughly a factor of two once attention is limited to statistically significant results, and the weight-for-age estimator is shown to be biased downward by the very mortality effect the study reports.

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2 Summary

Two questions, answered before the detail.

2.1 Does the deposited archive run?

No. None of its 15 scripts runs to the end, in either pass, and for fourteen of them the cause is the same: the deposit is written against the version of DeclareDesign that existed when the paper was published, and the package's interface changed at version 1.0.

The failures are not a matter of a deprecated name still working with a warning. `declare_estimand()`, `declare_population()` and `declare_reveal()` all still exist, and a script that used only those would run. What does not survive is the argument structure. `declare_sampling(n = 100)` and `declare_assignment(blocks = block, m = 10)` no longer describe a sampling or an assignment, `declare_estimator(model = lm_robust)` no longer names an estimation method, and `Z_cond_prob`, which the working example's inverse probability weighted estimator divides by, is no longer attached to the data at all. Fourteen of the fifteen stop at the first of these they reach.

The fifteenth is the experimental design script behind Figure 1, and it is the only one built entirely from `DesignLibrary` designers rather than from hand-written declarations, so the interface change does not touch it. It reaches the ten minute limit this harness gives each script and is recorded as unfinished, because it recomputes ten diagnoses at 5000 simulations first. Run separately at a reduced simulation count so that it gets past them, it stops at `write_rds(diagnosis_1, path = path_to_file_1)` with `argument "file" is missing, with no default`: `readr` renamed that argument, and the rename is an error rather than a warning. So it fails too, one step later than the others.

Three consequences are worth separating from the failure itself.

The deposit ships its own results, and it recomputes them anyway. Every analysis script computes a diagnosis, writes it to `rds_files/`, and then reads it straight back before printing, and those saved objects are in the deposit, one per diagnosis at the published 5000 simulations. Whether a script computes or merely loads is controlled by a flag, and the deposit's own driver sets it to recompute, which is what the passes here do as well. Both passes are reported below regardless: 0 of 15 scripts complete as shipped and 0 of 15 complete once those saved objects are removed. The distinction does not separate anything here only because nothing gets far enough for it to matter.

Those saved objects are what makes this paper checkable at all. They hold the numbers behind every published table at full precision, computed by the authors' own code under the packages of its day, and they are the source of the deposit's column in the ground truth below.

Running the deposit where it sits would damage it. The deposit contains 25 uncommented calls that write to disk, and its own driver script, `run_replication.R`, opens a sink into `output/` for each script before sourcing it and writes figures into `figure/`. Every file in `output/`, `figure/` and `rds_files/` is itself a deposited member, so running that driver in place would truncate part of the deposit before any analysis ran. Both passes here run in

scratch copies under the session's temporary directory and drive the scripts directly rather than through `run_replication.R`. The number of deposited files either pass actually modifies is 0, which is what a run that stops at its first declaration leaves behind, and the second verification step of `run_all.R` re-checks all 620 members against the manifest after everything has finished.

2.2 Does the maintained rewrite reproduce the paper?

Yes at the level of the analysis, and the arithmetic of the answer needs a word of explanation. Of 306 published claims drawn from the article and its appendix, 268 can be compared against the rewrite and 170 of them agree at the precision the article prints.

The 98 that do not have one main cause: 61 of them differ by a single unit in the last digit the article prints, and in 91 cases the deposit's own saved diagnosis reproduces the published value exactly. Every diagnosis in this paper is a Monte Carlo simulation of 5000 draws, so a proportion carries a simulation standard error of roughly 0.007 and a table printed to two decimals cannot be reproduced digit for digit except by drawing the same 5000 samples. The rewrite sets the deposit's own seed and the current DeclareDesign draws a different sample from it. Those rows carry `environment` below, and they are not a failure of either analysis.

The other 39 rows that carry a recorded fault are where the interest is, and they are listed individually further down. They include the claims the rewrite is not expected to answer at all, which are counted here rather than left in the unverifiable bucket without a reason.

Table 1: Published claims by type

Claim type	Claims	Comparable	Agree	Differ
pipeline	255	253	158	95
definitional	26	0	0	0
descriptive	15	15	12	3
structural	5	0	0	0
transcribed	5	0	0	0

`pipeline` claims are numbers the code must produce. `descriptive` claims are statements about shape, sign or order (that one estimator is less biased than another, that a line rises) which carry no printed value and are evaluated here as computed truth values. `definitional` claims are design parameters and conventions the article states, `structural` claims describe the document rather than a result, and `transcribed` claims are numbers copied from other papers.

Table 2: Disagreements between the rewrite and the article

Where the fault lies	Claims
environment	80
archive	14
paper_internal	5
unresolved	1

`environment` means a package moved under code that did not change. `archive` means the deposit cannot support the claim, through its code or its data. `paper_internal` means the article disagrees with its own tables. `unresolved` means the cause has not been established, which is recorded rather than guessed at.

The deposit's own saved results say something sharper from the other side. 240 of the 246 claims for which the deposit carries a comparable number agree with what the article printed, and every one of the 6 that do not is a sentence rather than a table cell.

Table 3: Every published value the deposit's own saved results do not reproduce

Float or section	Claim	Published	Deposit	Rewrite	Fault
Appendix S3.1.2	Percentage reduction in posterior standar...	40%	35%	35%	paper_in
Appendix S3.1.2	Percentage increase in bias under informa...	33%	1425%	1439%	paper_in
Appendix S2.1	Population average treatment effect of th...	0.406	0.409	0.410	unresolve
Appendix S4.7	Interact CBO has the highest RMSE of the ...	TRUE	FALSE	FALSE	paper_in
Appendix S4.5	Mean mortality estimand as stated in prose	0.006	-0.057	-0.057	paper_in
Appendix S4.5	Share of simulations returning a signific...	0.001	0.291	0.296	paper_in

3 Original Archive Reproducibility

Table 4: Every deposited script, run both ways

Script	Pass	Completed	Stopped at
1_simple_design.R	As shipped	no	Error: You appear to have used legacy d
1_simple_design.R	Stripped	no	Error: You appear to have used legacy d
2_design_census.R	As shipped	no	Error: You appear to have used legacy d
2_design_census.R	Stripped	no	Error: You appear to have used legacy d
3_10_discovery.R	As shipped	no	Error: You appear to have used legacy d
3_10_discovery.R	Stripped	no	Error: You appear to have used legacy d

3_1_survey_designs.R	As shipped	no	Error: You appear to have used legacy d
3_1_survey_designs.R	Stripped	no	Error: You appear to have used legacy d
3_2_Bayesian_descriptive_inference.R	As shipped	no	Error: You appear to have used legacy d
3_2_Bayesian_descriptive_inference.R	Stripped	no	Error: You appear to have used legacy d
3_3_process_tracing.R	As shipped	no	Error: You appear to have used legacy d
3_3_process_tracing.R	Stripped	no	Error: You appear to have used legacy d
3_4_QCA.R	As shipped	no	Error in file(file, ...) : argument "file" is
3_4_QCA.R	Stripped	no	Error in file(file, ...) : argument "file" is
3_5_nested.R	As shipped	no	Error in knitr::include_graphics("figure/
3_5_nested.R	Stripped	no	Error in knitr::include_graphics("figure/
3_6_model_based_estimands.R	As shipped	no	Error: You appear to have used legacy d
3_6_model_based_estimands.R	Stripped	no	Error: You appear to have used legacy d
3_7_matching.R	As shipped	no	Error: object 'bias' not found
3_7_matching.R	Stripped	no	Error: object 'bias' not found
3_8_regression_discontinuity.R	As shipped	no	Error in file(file, ...) : argument "file" is
3_8_regression_discontinuity.R	Stripped	no	Error in file(file, ...) : argument "file" is
3_9_experimental.R	As shipped	no	Did not finish within 600 seconds.
3_9_experimental.R	Stripped	no	Did not finish within 600 seconds.
4_B_and_S_replication.R	As shipped	no	Error: You appear to have used legacy d
4_B_and_S_replication.R	Stripped	no	Error: You appear to have used legacy d
appendix_code.R	As shipped	no	Error: You appear to have used legacy d
appendix_code.R	Stripped	no	Error: You appear to have used legacy d
table_3_declaration.R	As shipped	no	Error: You appear to have used legacy d
table_3_declaration.R	Stripped	no	Error: You appear to have used legacy d

The archive carries two seeds. `run_replication.R` sets `set.seed(90095)` before its loop, and every script it sources then sources `knitr_setup.R`, which sets `set.seed(90405)`. The second is therefore the seed each deposited script actually runs under, and it is the seed the rewrite uses.

What the deposit does supply is its own output. Because it saves every diagnosis at the published simulation count, the numbers behind every appendix table can be read from the deposit at full precision without running anything. `ground_truth/extract_archive_values.R` reads them, and they are the `value_script` column of the ground truth. That is what makes a three-way comparison possible here: the article's printed number, the deposit's saved number, and the rewrite's fresh computation.

4 Errata

The rewrite corrects two errors in the deposited code. Neither is an analytical choice; both are cases where the deposited code does not do what the surrounding text says it does.

The QCA truth table reads a column that does not exist. `3_4_QCA.R` passes `outcome = "Y_Obs"` to `truthTable()`, while the column the design creates is `Y_obs`. Under the current QCA package that spelling sends every call into the deposit's own `tryCatch` and returns `NO SOLUTION` for all of them, which would make both algorithms' failure rate 1.00. The appendix prints failure rates of 0.07 and 0.00, and prints a single draw of the estimates in which both algorithms return `B`, so the deposited spelling cannot be what produced the published table. The rewrite uses the column's own name.

The Bayesian design's sample size is not reachable by `redesign()`. `3_2_Bayesian_descriptive_inferen` writes the sample size as a literal inside `declare_sampling()` and then calls `redesign(bayesian_design, n = c(10, 100))`. The redesigned n = 100 design therefore samples ten units, exactly as the n = 10 design does. The published table shows this directly: its n = 10 and n = 100 rows are identical to two decimal places in all five columns. The rewrite names the sample size so that `redesign()` reaches it, which is what the published column headings say the table varies. A second table has the same shape and is treated differently below, because there the declaration needs no change and the current package reaches the parameter on its own.

---

5 What Moved Underneath the Code

Five changes in packages, none of them errors in the deposit, account for most of what the rewrite does differently. Each was found by running the code, not by reading a changelog.

The QCA package's notation for absence changed, and the diagnosand is a string comparison. The design's inquiry is the configuration `a*B`, written in the notation the article uses, where a lower case letter means the condition is absent. QCA 3.25 writes the same configuration as `~A*B`. The two diagnosands here are `mean(estimand == estimate)` and `mean(estimate == "NO SOLUTION")`, so the change of notation alone drives every correct rate to zero while leaving the failure rates untouched, and it does so silently: nothing errors, and the table still fills. The rewrite puts the estimator's answer back into the inquiry's notation before the two meet.

The two minimization algorithms the appendix compares now return the same answer. The design asks the classic Quine-McCluskey algorithm and the consistency cubes algorithm for the sufficient configuration, and the published tables show them differing: the classic algorithm fails to return a solution more often. Under QCA 3.25 the two return the identical solution in every one of the 5000 simulations, in all three of the appendix's QCA

tables, so the rewrite's consistency cubes column is a copy of its classic column. The deposit's saved diagnosis, computed under the QCA of its day, distinguishes them. Nothing errors here either.

`redesign()` now reaches a value a handler closes over. The nested mixed methods design is diagnosed across three decision thresholds. Two of them are read inside the body of a function passed as a handler. The published table varies with the first threshold and not with the other two: its eight rows take two distinct values, one for each setting of the large-N threshold. The current `redesign()` does reach all three, and the four rows at the high large-N threshold separate into a pair near 0.05 and a pair near 0.64, with the published value sitting between them. Running the same declaration with the large-N threshold high and the two SNA thresholds left at the defaults the deposit declares returns the four published rows at that setting to within simulation error, which is what identifies the published table as varying one threshold rather than three. The rewrite changes nothing in the declaration here, so this belongs to the package rather than to the deposit.

A neighbouring table redesigns on parameters that are in no version of the design. The same declaration is also diagnosed across the effort spent on model testing and model building, and those two parameters are used by sampling steps that the handler calls rather than by steps of the design itself. `redesign()` cannot reach them, and the current DeclareDesign says so in a warning as the rewrite runs. The four rows of that table are four simulations of one design, in the deposit as much as in the rewrite: the four published values differ by no more than simulation error, which is what that structure predicts. The appendix reads an ordering off them, that "increasing the effort devoted to testing decreases our chances of happening on the right model," and the ordering is not an effect of effort.

`QCA::simplify` is not exported, and was not exported when the paper was written either. The least squares answer strategy in the QCA appendix passes its predicted configurations to `QCA::simplify` inside a `try()`, so the call fails and the estimator returns `NO SOLUTION` on every simulation. The published table says exactly that: the least squares column is a failure rate of 1.00 and a correct rate of 0.00 in all four of its panels. The rewrite reproduces the published column, and it reproduces it for the same reason the deposit did.

A package loaded for one table breaks an unrelated design for the rest of the session. The QCA package loads `declared`, whose load hook writes over fourteen bindings in `base` and `stats`, among them `base::rbind.data.frame()`, which afterwards errors on any list argument that is not a data frame. `DesignLibrary::multi_arm_designer()`, which declares the three-arm design behind Figure 1, assembles its estimates with exactly that call. Figure 1 therefore cannot be diagnosed in a session where QCA has been loaded, and the failure names a base function far from the package that caused it.

Reaching QCA through its namespace does not avoid this: a load hook runs on load, so the first `QCA::` call installs the replacements just as `library(QCA)` would, and they last for the rest of the session. What protects Figure 1 is that it runs before the QCA appendix table, and `figure_1_experimental_designs.R` asserts that condition rather than relying on the

order of `run_all.R`. All fourteen were exercised on ordinary inputs before and after the load, and all fourteen return what the base versions return, `rbind.data.frame()` included when its arguments are data frames. The single behaviour that changes is `rbind.data.frame()` on an argument that is a plain list, which fails with `object 'xideclared' not found`, and that is the call the three-arm estimator makes.

---

6 Where the Article Disagrees With Itself

Four claims in the text are contradicted by the article's own tables or by both sets of code.

The power of the mortality estimator is printed twice, differently. The table in appendix section 4.5 gives a power of 0.29. The sentence below it reports that "only 0.1% of simulations returned a statistically significant result." The deposit's saved diagnosis agrees with the table.

The interaction strategy is not dominated. The appendix says that including the treatment-by-covariate interaction is "a strictly dominated strategy from the standpoint of reducing mean squared error: irrespective of whether CBO presence is correlated with health outcomes or imbalanced, the RMSE expected under this strategy is higher than under any other strategy." The article's own table prints a higher RMSE for ignoring the covariate than for interacting with it when CBO presence is correlated with outcomes. The deposit's saved diagnosis and the rewrite both agree with the table rather than with the sentence.

Neither percentage in the Bayesian comparison is reachable. The article states that informative priors "decrease the posterior standard deviation by 40% relative to the uniform priors, but increase the bias by 33%." Working from the deposit's own saved diagnosis gives a smaller reduction and an enormous increase, because the flat prior's bias is essentially zero, and the rewrite agrees with the deposit on both.

The exaggeration ratio is defined at one threshold and computed at another. The appendix defines it as "the ratio of the average absolute value of statistically significant estimates to the true effect, at the a = 0.05 level." The deposited diagnosand selects estimates on `p.value < 0.1`. The rewrite keeps the deposit's threshold, because it is the estimator the published table was computed with, and records the discrepancy rather than silently choosing one.

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7 Ground Truth Verification

The appendix numbers none of its tables, so this report names each by the section that prints it. The article's own floats keep their numbers. `ground_truth/published_claims.csv` is the transcription, hand-read from the published PDF and the published appendix, and it is the only file in the repository carrying a typed published value. Everything else is computed: `ground_truth/extract_archive_values.R` reads the deposit's saved results, `ground_truth/build_ground_truth.R` reads `maintained/output/`, and the comparison is made by printing the computed number to the precision the article printed and comparing digits.

The build stops rather than reporting a gap. A published float with no claim, a claim that should have a computed counterpart and finds none, and a disagreeing row with no recorded fault all halt it.

The transcription was checked by extracting every numeric token from the published article and the published appendix and looking for tokens the ground truth does not carry. What remains after that pass is section numbers, page footers, citation years, and the model parameters the appendix prints inside its code blocks, which the rewrite's declarations carry verbatim and which the claims file checks by name where the surrounding prose states them. One published numeral is genuinely uncovered: the appendix's coding rules give the standard deviation of the working example's outcome as 1.01, no deposited script computes a standard deviation of an outcome and no file in `maintained/output/` carries one, so there is nothing to compare it against on either side. The average treatment effect stated in the same sentence is covered, and neither the deposit nor the rewrite reproduces it.

Table 5: Coverage of every published float

Float or section	Claims	Comparable	Agree
Table 1	7	0	0
Table 3	4	0	0
Table 4	1	0	0
Table 2	16	13	13
Figure 1	5	2	2
Appendix S3.2.1a	22	18	11
Appendix S3.2.2e	25	24	18
Appendix S4.5	18	11	7
Appendix S4.6	14	11	5
Appendix S4.7	26	25	16
Appendix S3.1.2	27	27	16
Appendix S3.2.2d	17	17	7
Figure 2	2	2	2

Appendix S2.1	19	15	5
Appendix S3.2.1b	3	3	2
Appendix S3.2.2c	6	6	1
Appendix S3.2.3a	25	25	25
Appendix S3.2.3b	5	5	0
Appendix S3.2.5	3	3	3
Appendix S3.3	17	17	11
Appendix S1	3	3	3
Appendix S3.1.1	1	1	1
Appendix S3.2.2a	6	6	6
Appendix S3.2.2b	2	2	2
Appendix S3.2.3c	8	8	0
Appendix S3.2.4	20	20	10
Appendix S3.2.6	4	4	4

7.1 The disagreements

A disagreement of one unit in the last place the article prints is a different kind of thing from a disagreement of ten, and the two are separated here by measuring the gap rather than by asserting it.

Everything larger than one unit is listed in full below, with its reason recorded in the `notes` column of the ground truth. Several of them are still differences of simulation draw: the nested tables are printed to four decimals, where the simulation error of 5000 draws is worth about seventy units in the last place.

Table 6: Every disagreement of more than one unit in the last printed place

Float or section	Claim	Published	Deposit	Rew
Table 2	Table 2 reports 0 of the 30 admitted tool...	0	-	-
Appendix S3.1.2	Percentage reduction in posterior standar...	40%	35%	35%
Appendix S3.1.2	Percentage increase in bias under informa...	33%	1425%	1439
Appendix S2.1	Population average treatment effect of th...	0.406	0.409	0.41
Appendix S2.1	Standard deviation of the working example...	1.01	-	-
Appendix S3.2.2c	Consistency Cubes has a lower failure rat...	TRUE	TRUE	FAL
Appendix S3.2.3b	More effort on theory building and less o...	TRUE	TRUE	FAL
Appendix S4.7	Interact CBO has the highest RMSE of the ...	TRUE	FALSE	FAL
Appendix S2.1	Appendix S2.1, IPW-BFE: power	0.72	0.72	0.74
Appendix S3.1.2	Appendix S3.1.2, 100|flat priors: mean_sd	0.14	0.14	0.05
Appendix S3.1.2	Appendix S3.1.2, 100|flat priors: sd_shift	-0.15	-0.15	-0.24

Appendix S3.1.2	Appendix S3.1.2, 100|informative priors: ...	0.49	0.49	0.47
Appendix S3.1.2	Appendix S3.1.2, 100|informative priors: ...	0.09	0.09	0.05
Appendix S3.1.2	Appendix S3.1.2, 100|informative priors: ...	0.03	0.03	0.01
Appendix S3.1.2	Appendix S3.1.2, 100|informative priors: ...	-0.01	-0.01	-0.03
Appendix S3.1.2	Appendix S3.1.2, 100|informative priors: ...	-0.02	-0.02	-0.06
Appendix S3.2.1a	Appendix S3.2.1a, Update from both clues:...	0.51	0.51	0.49
Appendix S3.2.2c	Appendix S3.2.2c, Consistency Cubes: fail...	0.00	0.00	0.07
Appendix S3.2.2d	Appendix S3.2.2d, 0|5|Consistency Cubes: ...	0.01	0.01	0.12
Appendix S3.2.2d	Appendix S3.2.2d, 0.05|5|Consistency Cube...	0.01	0.01	0.15
Appendix S3.2.2e	Appendix S3.2.2e, 0|3|Classic Quine-McClu...	0.32	0.32	0.30
Appendix S3.2.2e	Appendix S3.2.2e, 0|3|Consistency Cubes: ...	0.05	0.05	0.30
Appendix S3.2.2e	Appendix S3.2.2e, 0.2|3|Consistency Cubes...	0.10	0.10	0.44
Appendix S3.2.3b	Share of simulations reaching the New Mod...	0.2780	0.2780	0.269
Appendix S3.2.3b	Share of simulations reaching the New Mod...	0.2858	0.2858	0.273
Appendix S3.2.3b	Share of simulations reaching the New Mod...	0.2782	0.2782	0.287
Appendix S3.2.3b	Share of simulations reaching the New Mod...	0.2752	0.2752	0.274
Appendix S3.2.3c	Share of simulations reaching the New Mod...	0.6424	0.6424	0.629
Appendix S3.2.3c	Share of simulations reaching the New Mod...	0.6270	0.6270	0.627
Appendix S3.2.3c	Share of simulations reaching the New Mod...	0.6244	0.6244	0.630
Appendix S3.2.3c	Share of simulations reaching the New Mod...	0.6242	0.6242	0.638
Appendix S3.2.3c	Share of simulations reaching the New Mod...	0.2840	0.2840	0.053
Appendix S3.2.3c	Share of simulations reaching the New Mod...	0.2748	0.2748	0.635
Appendix S3.2.3c	Share of simulations reaching the New Mod...	0.2740	0.2740	0.050
Appendix S3.2.3c	Share of simulations reaching the New Mod...	0.2738	0.2738	0.635
Appendix S4.5	Mean mortality estimand as stated in prose	0.006	-0.057	-0.05
Appendix S4.5	Share of simulations returning a signific...	0.001	0.291	0.296
Appendix S4.6	Appendix S4.6, bjorkman_svensson_design|M...	0.28	0.28	0.30
Appendix S4.6	Appendix S4.6, RPP_replication|Weight est...	2.02	2.02	1.99

The remainder differ by exactly one unit in the last printed place and are counted by the float they belong to.

Table 7: Claims differing only in the last printed digit

Float or section	Claims
Appendix S2.1	8
Appendix S3.1.2	2
Appendix S3.2.1a	6
Appendix S3.2.1b	1
Appendix S3.2.2c	3

Appendix S3.2.2d	8
Appendix S3.2.2e	3
Appendix S3.2.4	10
Appendix S3.3	6
Appendix S4.5	2
Appendix S4.6	4
Appendix S4.7	8

`maintained/in_text_claims.R` is the second instrument. It walks the article and its appendix in reading order, carrying the sentence each claim comes from in a comment beside the code, and computes the number that backs it from `maintained/output/` directly. It never reads the ground truth. The two therefore reach every claimed number by separate paths, each doing its own selection and its own rounding, and the ground truth build runs the second one and compares line by line. A disagreement between them halts the build, because one of the two would be wrong.

---

8 Maintained Rewrite

Table 8: Scripts in maintained/

Script	Role
bs_design.R	The Bjorkman and Svensson declaration, sourced by three scripts
figure_1_experimental_designs.R	Article figure
figure_2_bs_sampling_distributions.R	Article figure
helpers.R	Packages, paths and the shared diagnosand writer
in_text_claims.R	Prints every published claim beside its computed counterpart
table_2_tool_census.R	Article table
table_3_declaration.R	Article table
table_a1_simple_design.R	Appendix table
table_a12_nested_analysis.R	Appendix table
table_a15_model_based_estimands.R	Appendix table
table_a16_matching.R	Appendix table
table_a17_regression_discontinuity.R	Appendix table
table_a18_discovery.R	Appendix table
table_a19_bs_diagnosis.R	Appendix table
table_a2_working_example_census.R	Appendix table

table_a20_bs_exaggeration.R	Appendix table
table_a21_bs_covariate_adjustment.R	Appendix table
table_a3_survey_design.R	Appendix table
table_a4_bayesian_descriptive.R	Appendix table
table_a5_process_tracing.R	Appendix table
table_a7_qca.R	Appendix table

Every diagnosis runs at the published 5000 simulations and 5000 bootstrap replicates, set once in `helpers.R` and read from `knitr_setup.R` in the deposit. `diagnose_design()` spreads its simulations over the `future` package, and it draws the same simulations however that work is divided, which is checked under the verification heading below.

The rewrite uses the current DeclareDesign interface throughout: `declare_inquiry()` for `declare_estimand()`, `declare_sampling(S = ..., filter = S == 1)` for `declare_sampling(n = ...)`, `declare_assignment(Z = block_ra(...))` for the old block and cluster arguments, `.method =` for `model =`, and `declare_measurement(Y = reveal_outcomes(...))` for `declare_reveal()`. Where the old interface supplied something automatically, the rewrite computes it explicitly: the working example's assignment probabilities come from `obtain_condition_probabilities()` over the same block sizes that generate the assignment, so the inverse probability weighted estimator the deposit could no longer form is available again.

---

9 Figure Verification

Neither figure prints a number, so each is verified against the estimates it is drawn from, and those estimates are written to a CSV beside the image.

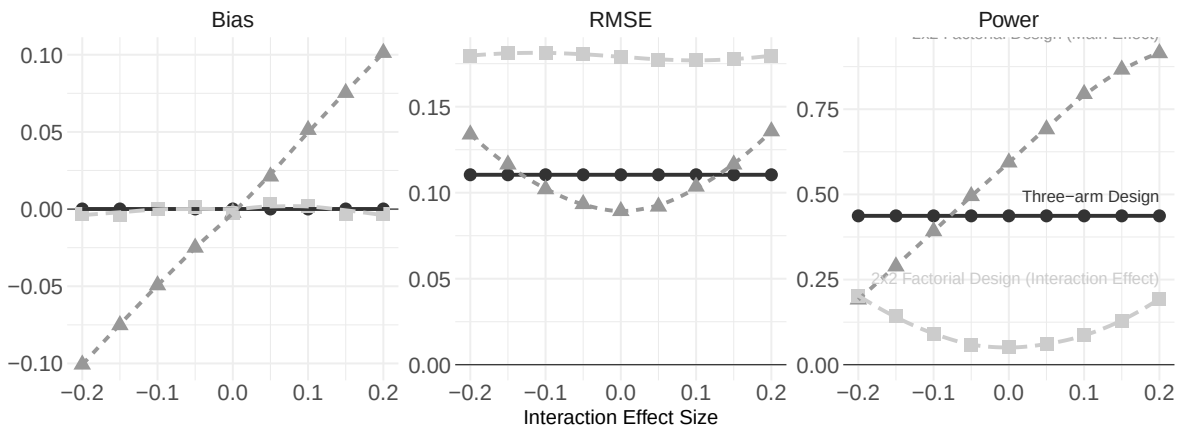


Figure 1: Article Figure 1: diagnoses of the factorial and three-arm designs

The article says of Figure 1 that neither design is biased when the true interaction is zero, and that the factorial design's bias grows as the interaction grows. The first is a number: the largest absolute main effect bias at an interaction of zero is 0.00. The second is a shape, and the factorial main effect bias rises across all nine interaction sizes.

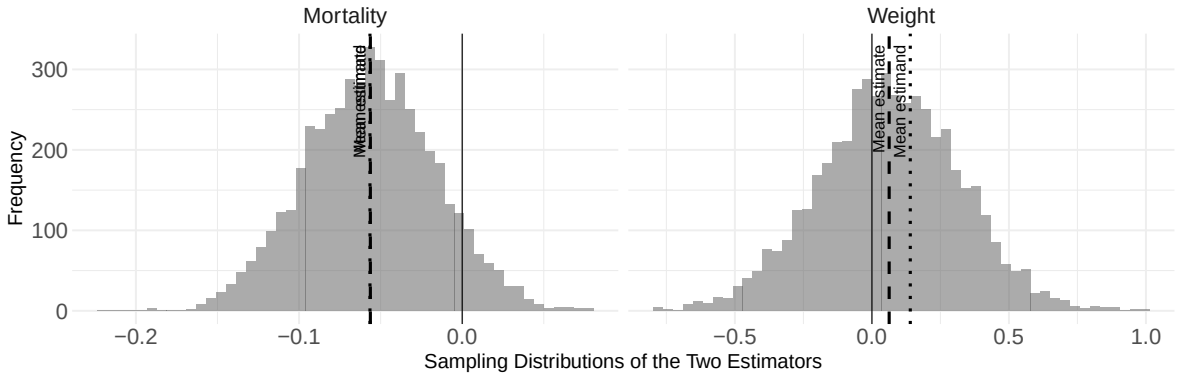


Figure 2: Article Figure 2: sampling distributions of the two Bjorkman and Svensson estimators

Figure 2 plots the same simulations that appendix section 4.5 summarises, so the figure and that table cannot disagree: `table_a19_bs_diagnosis.R` writes both the diagnosands and the per-simulation estimates, and the figure script reads the second of those back. The mean plotted mortality estimate is -0.06 and the mean plotted weight estimate is 0.06, matching the two cells the appendix table prints.

10 Maintained Rewrite Verification

Every script sets the deposit's seed before it declares anything, and every diagnosis is seeded, so a second run reproduces the first exactly. `run_all.R` was run end to end three times from clean sessions. Every CSV, TeX and PNG file came back byte-identical each time, as did the ground truth and the record of running the deposited code; only the two figure PDFs differ, because a PDF records the time it was written. The diagnoses also return the same numbers under a sequential plan and under forking at two and four workers, bootstrap standard errors included, which was measured rather than assumed.

`maintained/output/` is committed. A reader who runs `source("run_all.R")` and then `git status --porcelain maintained/output/` should see the two figure PDFs and nothing else.

The deposit is verified twice per run. `download_original.R` sources `verify_original.R` before anything else, which is a precondition, and `run_all.R` sources `verify_original.R` again as its last step, which is the demonstration. The second pass is deliberately not a re-run of `download_original.R`: that script rebuilds `original_extracted/` from the zip and would erase exactly the damage the check exists to find.

The deposited archive's MD5 is `f49430c4bf84c3dc375198ae7bd8f6ce`, which is also the checksum Dataverse publishes for it, over 41,188,788 bytes. A checksum taken of any file this repository writes would differ on every run, so none is quoted.

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11 R Environment

Table 9: Key package versions

Package	Version
DeclareDesign	1.1.1
DesignLibrary	0.1.10
estimatr	1.0.6
QCA	3.25
Matching	4.10.15
tidyverse	2.0.0
future	1.70.0
readxl	1.5.0
broom	1.0.13
knitr	1.51

kableExtra	1.4.0
here	1.0.2

R version: R version 4.6.0 (2026-04-24)